zihaodeng@umass.edu

Zihao Deng

530-761-6969

Multi-robot systems, Robot adaptation and learning

Education

University of Massachusetts, Amherst — Amherst, MA
Ph.D in Computer Science — *Feb 2025 – Present*

University of Massachusetts, Amherst — Amherst, MA
M.S. in Computer Science — *Feb 2023 – Dec 2024*

University of California, Davis — Davis, CA
B.S. in Computer Science — *Sep 2018 – Jun 2022*

Publications

1. **Zihao Deng**, Qianhuang Li, Maggie Wigness, John Rogers, Peng Gao, and Hao Zhang, "CoM-VLA: Coordinated Multi-Robot Vision-Language-Action Model for Decentralized Formation Reconfiguration", *Under Review.*

2. **Zihao Deng**, Peng Gao, Qianhuang Li, Maggie Wigness, John Rogers, Donghyun Kim, and Hao Zhang, "ReDiG: Reinforced Diffusion on Graphs for Decentralized Coordinated Multi-Robot Navigation with Smooth Formation Adaptation", *Under Review.*

3. Qianhuang Li, **Zihao Deng**, Oscar Youngquist, Maggie Wigness, John Rogers, Peng Gao, Hao Zhang, "Physics-Informed Intent-Aware Human-Robot Collaborative Transportation With Forward-Looking Adaptation", *Under Review.*

4. Qianhuang Li, **Zihao Deng**, Maggie Wigness, John Rogers, Hao Zhang, and Donghyun Kim, " Constrained Subswarm Flow for Decentralized Multi-Robot Planning with Emergent Split-Merge", *Under Review.*

5. **Zihao Deng**, Qianhuang Li, Maggie Wigness, John Rogers, Donghyun Kim, and Hao Zhang, "Collaborative Planning for UAV-UGV Teams under Operational Constraints via Masked Error-bounded Neuro-Symbolic Diffusion", *Conference on Robot Learning (CoRL)*, 2026. **Spotlight** [Paper] [Page]

6. Williard Joshua Jose, Yuhao Li, **Zihao Deng**, and Hao Zhang, " Adapting Execution-Time Objectives for Multi-Robot Policies via Collaborative Flow Policy Guidance", *Robotics: Science and Systems (RSS)*, 2026. [Paper] [Supplementary]

7. **Zihao Deng**, Qianhuang Li, Peng Gao, Maggie Wigness, John Rogers, Donghyun Kim, and Hao Zhang, "Collaborative Planning with Concurrent Synchronization for Operationally Constrained UAV-UGV Teams", *IEEE International Conference on Robotics and Automation (ICRA)*, 2026. [Paper] [Page]

8. **Zihao Deng**, Peng Gao, Williard Joshua Jose, Maggie Wigness, John Rogers, Brian Reily, Christopher Reardon, and Hao Zhang, "Subteaming and Adaptive Formation Control for Coordinated Multi-Robot Navigation", *Conference on Robot Learning (CoRL)*, 2025. [Paper] [Page]

9. **Zihao Deng**, Peng Gao, Williard Joshua Jose, Christopher Reardon, Maggie Wigness, John Rogers, and Hao Zhang, "Coordinated Multi-Robot Navigation with Formation Adaptation", *IEEE International Conference on Robotics and Automation (ICRA)*, 2025. [Paper] [Page]

Research Experience

University of Massachusetts, Amherst — Amherst, MA
Research Assistant — *June 2023 – Present*

- **Adaptive coordinated navigation:** Developed learning-based methods for coordinated multi-robot navigation and environment-aware adaptive formation control (ARL).
- **Heterogeneous collaborative planning:** Developed neuro-symbolic methods for collaborative planning of operationally constrained UAV–UGV teams (ARL).

Massachusetts Institute of Technology — Boston, MA(Remote)
Online Research Assistant — *Sep 2021 – Dec 2021*

- **Medical image analysis:** Developed deep learning models for automated skin cancer assessment from medical images.

Awards

1. **Robert and Deanna Hagerty Robotics Scholarship**, University of Massachusetts Amherst, Amherst, MA, USA, October 2026.

   Competitive scholarship awarded to outstanding students for excellence in robotics research.

2. **Travel Award Grant**, IEEE International Conference on Robotics and Automation (ICRA), Atlanta, GA, USA, May 2025.

Invited talk

1. **Coordinated and Collaborative Multi-Robot System**
   - North Carolina State University, 2025
   - University of Massachusetts, Amherst, 2025
2. **Robot Operating System 2 (ROS 2) Tutorial**
   - University of Massachusetts Amherst, 2026

Academia Service

Reviewer:

1. Robotics: Science and Systems (RSS), 2026.
2. Conference on Robot Learning (CoRL), 2025-present.
3. IEEE Robotics and Automation Letters (RAL), 2024-present.
4. IEEE International Conference on Robotics and Automation (ICRA), 2024-present.
5. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2024-present.
6. Neural Information Processing Systems (NeurIPS), 2026.

Outreach & Service

- Manning CICS CSL Ribbon Cutting – September 2026
- UMass Discovery Camp, Department of Recreation & Wellbeing – August 2026
- UMass Pre-College Entrepreneurship Program – July 2026
- NEMS Workshop, UMass Amherst – June 2026
- UMass Amherst Founders Day Robotics Showcase – April 2026
- Graduate Women in Business (GWIB) Tech-Fusion Robotics Lab Visit – December 2025
- Manning CICS Computer Science Laboratories Open House – November 2025
- Manning CICS COOL / Springfield Middle School Robotics Demonstration – October 2025
- Manning CICS "Come Home to Your College" – August 2025
- UMass Isenberg Entrepreneurship Game Changer Program – July 2025
- Alfred G. Zanetti Montessori School / Massenberg STEM Institute Robotics Lab Visit – October 2024
- UMass Amherst CICS Turing Summer Program – July 2023

Skills

Programming Languages: Python, C/C++, MATLAB, R, Rust
Machine Learning: PyTorch, TorchRL, PyTorch Geometric, TensorFlow, NumPy, Pandas, Scikit-learn, SciPy
Robotics Frameworks: ROS 1 (Melodic, Noetic), ROS 2 (Foxy, Humble, Jazzy)
Simulation: Isaac Lab, Isaac Sim, Gazebo, MuJoCo, Pygame